Lab Report - Mobile Charger

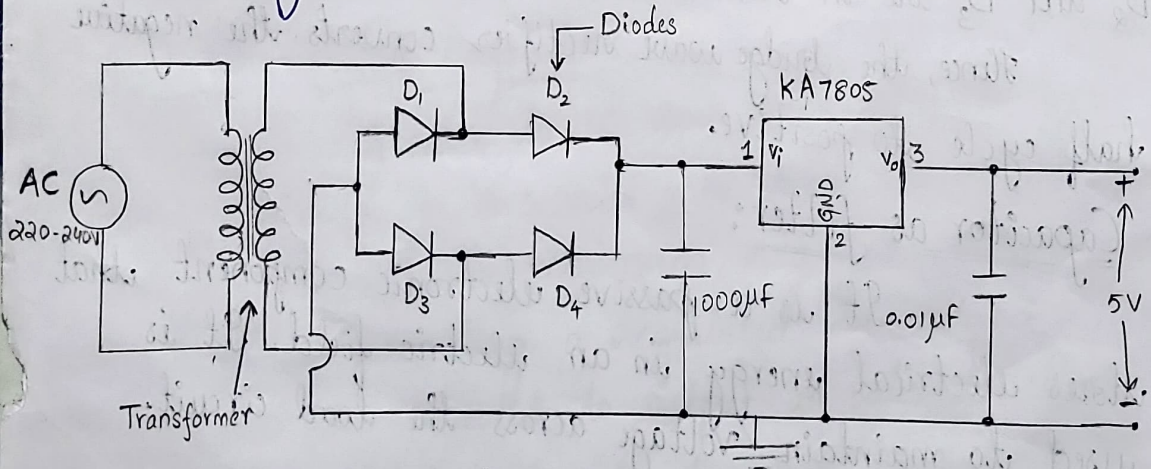
Aim:

The objective of the project is to construct and demonstrate a mobile Charger.

Components:-

1) Diodes
2) 1000μF capacitor
3) 0.01μF capacitor
4) Transformer (230V → 12V)
5) 7805 IC Regulator
6) USB Connector
7) Bread board.

Circuit Diagram:-



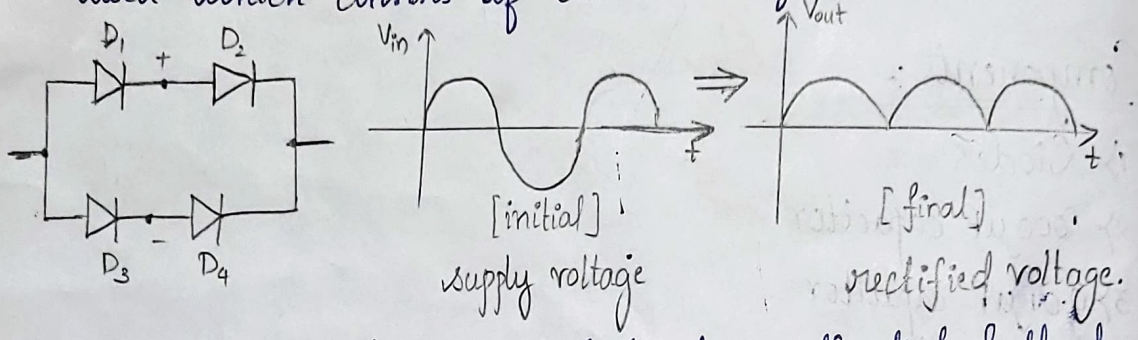
Working Procedure of the Circuit:

(i) Transformer:-

In this experimental setup a step down transformer is used to reduce the input AC voltage which supplied to the circuit from 230V to 12V.

(ii) Diodes:

It is a basic electrical component which is used to allow the current to flow in one direction. In this experimental setup, a bridge wave rectifier is used which consists of combination of the diodes.



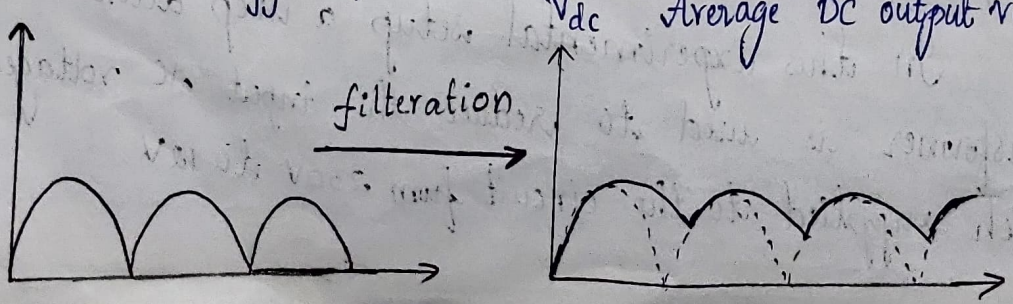
These combination of diodes during the first half of cycle i.e, when voltage flow is in positive direction; D_2 and D_3 are in forward bias & D_1 and D_4 are in reverse bias. During the second half of cycle i.e, when voltage flow is in negative direction; D_1 and D_4 are in forward bias & D_2 and D_3 are in reverse bias.

Hence, the bridge wave rectifier converts the negative half cycle to positive.

(iii) Capacitor as filter:

It is a passive electronic component that stores electrical energy in an electric field. It is used to maintain voltage across the load circuit. It helps to low the ripple factor.

$$\text{Ripple factor} = \frac{V_{ac}}{V_{dc}} = \frac{\text{Rectified RMS voltage}}{\text{Average DC output voltage}}$$



Consider, value of filter capacitor:

$$C = \frac{I \times t}{V}$$

where,

I = maximum output current (500mA for the given transformer).

C = capacitance.

$t = \frac{1}{f}$; Voltage before rectification had frequency of 50Hz & after rectification it had 100Hz.

$$\therefore t = \frac{1}{100\ s^{-1}} = 10\ ms$$

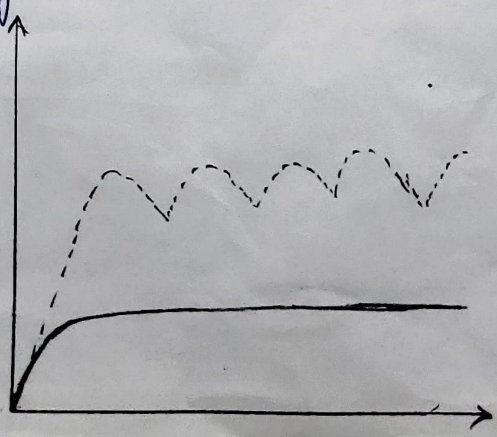
The KA-7805 regulator operates between 7V and 3.2V and it returns the output voltage in the range of 4.8 V - 5.2V.

$$\therefore V = 5\ V$$

$$C = \frac{500 \times 10^{-3} \times 10 \times 10^{-3}}{5} = 1000\ \mu F.$$

(iv) KA-7805 Voltage Regulator:

In order to IC work properly, the input voltage should be 2-3 volts more than the output voltage value, i.e., the minimum required voltage is 7V. It helps in providing a regulated DC voltage. A 0.01 μf is connected to output in order to eliminate the noise, produced due to transient changes in the voltage.



Precautions:

① Be careful while making the circuit as 230V AC mains is being used.

② As the capacitor is bipolar, ensure that the positive and negative terminals are placed properly while making the connections.

Results:

1. Expected output voltage :- 5V
   Observed output voltage :- 4.98V
2. Input AC voltage : expected :- 230V
   observed :- 234V
3. Output of transformer : expected = 12V
   observed = 13.3V

Conclusion:

The experiment successfully demonstrated its objectives by designing and constructing a reliable mobile charger with regulated 5V output. The results suggests that charger is suitable for practical user and provides stable output for mobile devices.